

Multimedia

• Multimedia

↳ Multimedia is the field of computing that involves the integration of multiple forms of media. It combines the five core elements: Text, Graphics, Audio, Video, and Animation.

The goal is to deliver information or entertainment in an engaging and multi-dimensional way. It is delivered using electronic or digital means (e.g., Websites, apps, games, presentations).

GSS Notes

When the user controls the flow of information, it is called interactive multimedia.

1) Text

↳ Text is the most fundamental element in any multimedia application.

> Definition: Text consists of characters, words, sentences, and paragraphs that communicate specific information.

> Role in Multimedia

- It provides the foundation for structure (heading, labels), navigation (hyperlinks, menus), and detailed content (narration scripts, captions).

> Types:

- Plain Text (ASCII): Unformatted text, simple character data.
- Formatted Text: Text with specific fonts, sizes, styles (bold, italic), and colors.
- Hypertext: Text that contains links to other documents or media elements.

eg: On a Webpage, the sentence:

÷ Learn more about photosynthesis."

Here, the word photosynthesis might be clickable to take you to a page explaining it. That's hypertext.

- Considerations: Readability is key. Designers must choose appropriate fonts (typefaces), sizes, and colors to ensure legibility on the screen.

2. Images (Graphics)

↳ Images are static visual representations

that convey information quickly and ~~add~~ add more charm.

> Definition - Visual data formats including photographs, drawings, charts and diagrams.

> Role in Multimedia

- They are used to illustrate complex ideas, provide content, create background textures, and enhance the visual quality of the presentation.

> Types:

- Raster (Bitmap) Graphics

- Raster graphics are images made up of a grid of tiny pixels, where each pixel has its own color value.

* Key characteristics

- Composed of pixels (tiny squares of color).
- Resolution-dependent (if you enlarge it too much, it becomes pixelated).
- Best for detailed and complex images, like photographs.
- Common file formats: JPEG, PNG, GIF, BMP, TIFF.

• Vector Graphics

- Vector graphics are images made up of mathematical equations defining shapes, lines, curves, and colors.

* Key characteristics:

- Composed of paths (lines and curves) instead of pixels.
- Resolution-independent (can be scaled infinitely without losing quality).
- Ideal for logos, icons, and illustrations.
- Common file formats: SVG, AI, EPS, PDF.

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- > Considerations: Image resolution, file size, and the balance between quality and loading speed (compression) are crucial.

③ Audio

↳ Audio refers to all sound elements used in a multimedia application.

- > Definition - Digitized sound, including speech, music, and sound effects.

- > Role in Multimedia - It can set the mood

(background music), provide instructions (narration / voiceover), and give feedback (interface sound effects). It significantly increases engagement and comprehension.

> Types:

- Speech

↳ Narration, dialogue, or voiceover.

- Music

↳ Background scores or theme songs.

- Sound Effects

↳ Environmental sounds, button clicks, or specific event cues.

> Considerations:

- Sound quality (sampling rate and bit depth), file format (MP3, WAV, AAC), and synchronization with visual elements are important for a polished product.

④ Video

↳ Video is a dynamic element that presents a continuous stream of visual and auditory information.

> Definition

- A sequence of still images (frames) played

back rapidly (typically 24 to 30 frames per second) to create the illusion of motion, often coupled with synchronized audio.

> Role in Multimedia

- It is the best element for demonstrating real-world actions, showing complex processes, or telling a compelling story realistically.

> Analog vs Digital

- Multimedia primarily uses Digital Video, which requires compression (codecs like H.264 or HEVC) to manage large file sizes.

> Considerations

- Frame rate, aspect ratio, resolution (e.g. 720p, 1080p, 4K), and the choice of compression codec are critical for video quality and streaming performance.

⑤ Animation

↳ Animation is a time-based visual element that depicts motion through sequential display of frames.

> Definition

- The process of making a static image

appears to move, primarily used for illustration where real video footage might be impossible or too complex.

> Role in Multimedia

- It's excellent for illustrating abstract concepts, showing step-by-step processes, creating smooth transitions, or bringing graphics to life (e.g. an animated logo).

> Types:

• 2D Animation

- Creates motion in two-dimensional plane (e.g. cartoons, motion graphics).

• 3D Animation

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- Creates virtual objects that can be rotated and viewed from any angle in a three-dimensional space (e.g. architectural walkthroughs, medical modeling).

> Considerations

- Smoothness of motion (frame interpolation), complexity of rendering (especially for 3D), and file size (e.g. GIFs vs video formats) are key design factors.

Write down 5-5 advantages and disadvantages of multimedia.

→ Any 5 advantages of multimedia are:

- i) Engaging - Combines text, sound, and visuals to hold attention.
- ii) Easy Understanding - Explains difficult idea more clearly.
- iii) Better Memory - Visual and audio content is easier to remember.
- iv) Interactive - Users can participate and learn actively.
- v) Wide use - Helpful in education, business, and entertainment.

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→ Any 5 disadvantages of multimedia

- i) costly - Needs money for software and equipment.
- ii) Technical issues - Errors can disturb work or learning.
- iii) High storage - Files take more space and data.
- iv) Hardware dependent - Requires good devices to run smoothly.
- v) Distracting - Too much media can reduce focus.

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